

ShopFlow AWS Project

ElastiCache Redis Design

Managed Redis 7.x - Session Store - Cache - Rate Limiter

Module 7 of 18

Field	Value
Version	1.0
Date	June 2026
Region	ap-south-1 (Mumbai)
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Application	ShopVerse -- Spring Boot 3.2.5 / Java 21

1. Overview

Amazon ElastiCache provides a managed Redis 7.1 cluster for ShopVerse. AWS handles patching, backups, and recovery. ShopVerse uses Redis for eight distinct purposes. The cluster resides in private subnets with TLS in-transit encryption and an AUTH token stored in Secrets Manager.

2. ShopVerse Redis Usage Map

#	Use Case	Key Pattern	TTL	Data Structure	Source Class
1	JWT Session Validation	session:{token}	24h	String	RedisSessionStore.java
2	Product Cache	products::{id}	30m	String (JSON)	CachedProductService.java
3	Order Cache	orders::{id}	5m	String (JSON)	OrderCacheService.java
4	Rate Limiter	rate:{ip}	1m	Counter (INCR)	RedisSessionStore.java
5	Shopping Cart	cart:{userId}	Session	Hash	Cart service
6	Distributed Locks	lock:{resource}	5s	String (NX)	DistributedLockService.java
7	Pub/Sub Low-Stock Alerts	low-stock-alerts	N/A	Channel	RedisPubSubPublisher.java
8	Recently Viewed Products	recent:{userId}	7d	List	Product service

3. Cluster Configuration

Setting	Value	Reason
Engine	Redis 7.1	Latest stable with enhanced I/O multiplexing
Node Type	cache.t3.micro	Free-tier eligible (750 hrs/month)
Nodes	1 (single-node)	Dev -- no replicas needed
Auth Token	32-char random	From Secrets Manager -- alphanumeric only
Transit Encryption	TLS enabled	Required when AUTH is used
At-Rest Encryption	AES-256	For RDB/AOF snapshots on disk
Snapshot Retention	3 days	Daily snapshots at 03:00-04:00 UTC
Eviction Policy	allkeys-lru	Evict least-recently-used from all keys

4. Spring Boot Connection

```
spring.data.redis.host=<REDIS_HOST>
spring.data.redis.port=6379
spring.data.redis.password=<REDIS_AUTH_TOKEN>
```

```
spring.data.redis.ssl.enabled=true  
spring.data.redis.lettuce.pool.max-active=16
```

5. Cost Estimate

Item	Monthly Cost
cache.t3.micro single node	\$11.52
3-day snapshot storage	~\$0.10
Total	~\$11.62 / month